

Unit ID: 679b1daeaaf593b2823f375d

Unit 3 - Conditional Statements

PDF Documents

PDF Document 1: 1738218924156-Chapter 8 - Unit 3 - Conditional Statements.pdf

This PDF document is included in the unit materials.

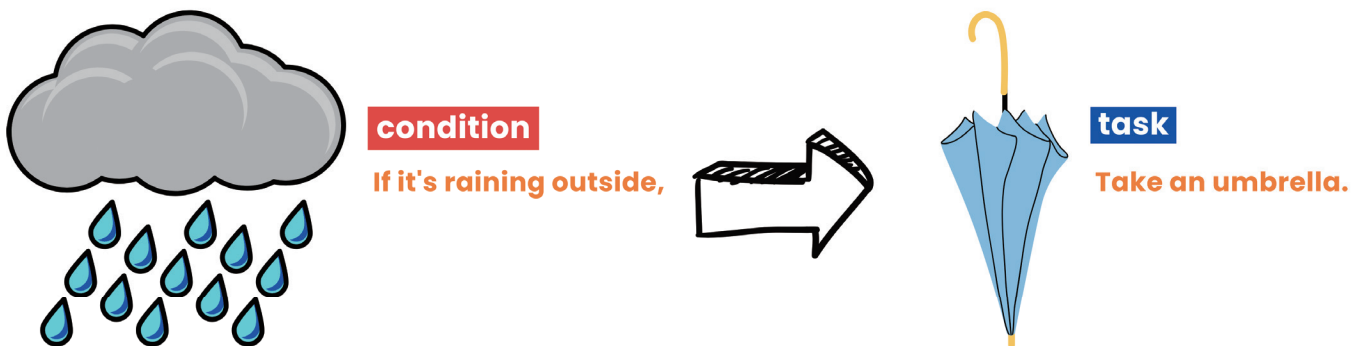
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UNIT 3 CONDITIONAL STATEMENTS

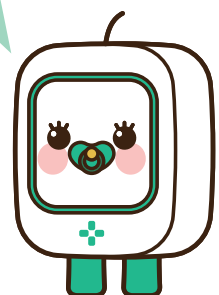
IF STATEMENT

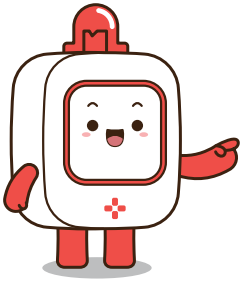
Think of an "if statement" like a rule or a condition in a game. It's like saying, "If something is true, then do something."



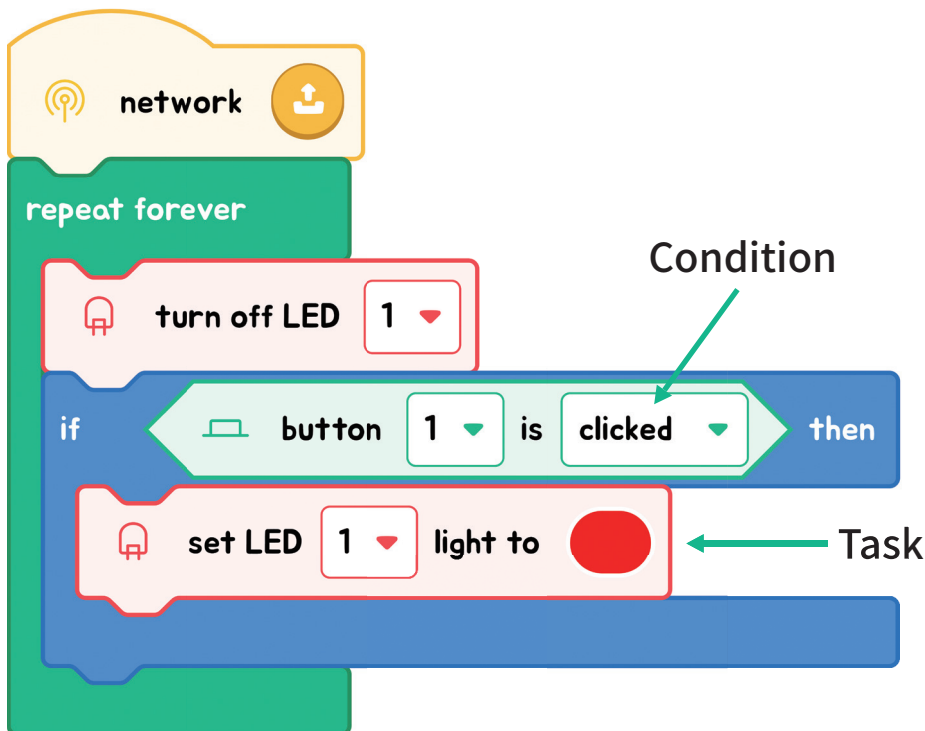
Imagine you have to decide whether to bring an umbrella outside. The rule could be: "If it's raining, take an umbrella. If it's not raining, you don't need to take an umbrella."

Can you find another example where we can use the "if" statement in our daily life?





Here's a simple explanation and example:
Let's create the code like below and upload it!



- 1** Condition Checking: First, the program checks a specific condition. For example, it might check if button is clicked
- 2** Execution if True: If the condition is true, the block of code inside the "if" statement is executed. This block contains the tasks that should be performed if the condition is true.

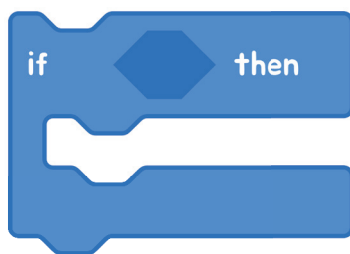


Conditional Statement1

Fill in the blanks with "condition," "false," and "true" as appropriate for the following statement:



Activity When the Button is clicked, the LED lights up in red. If not, LED is off.



Condition

False

True

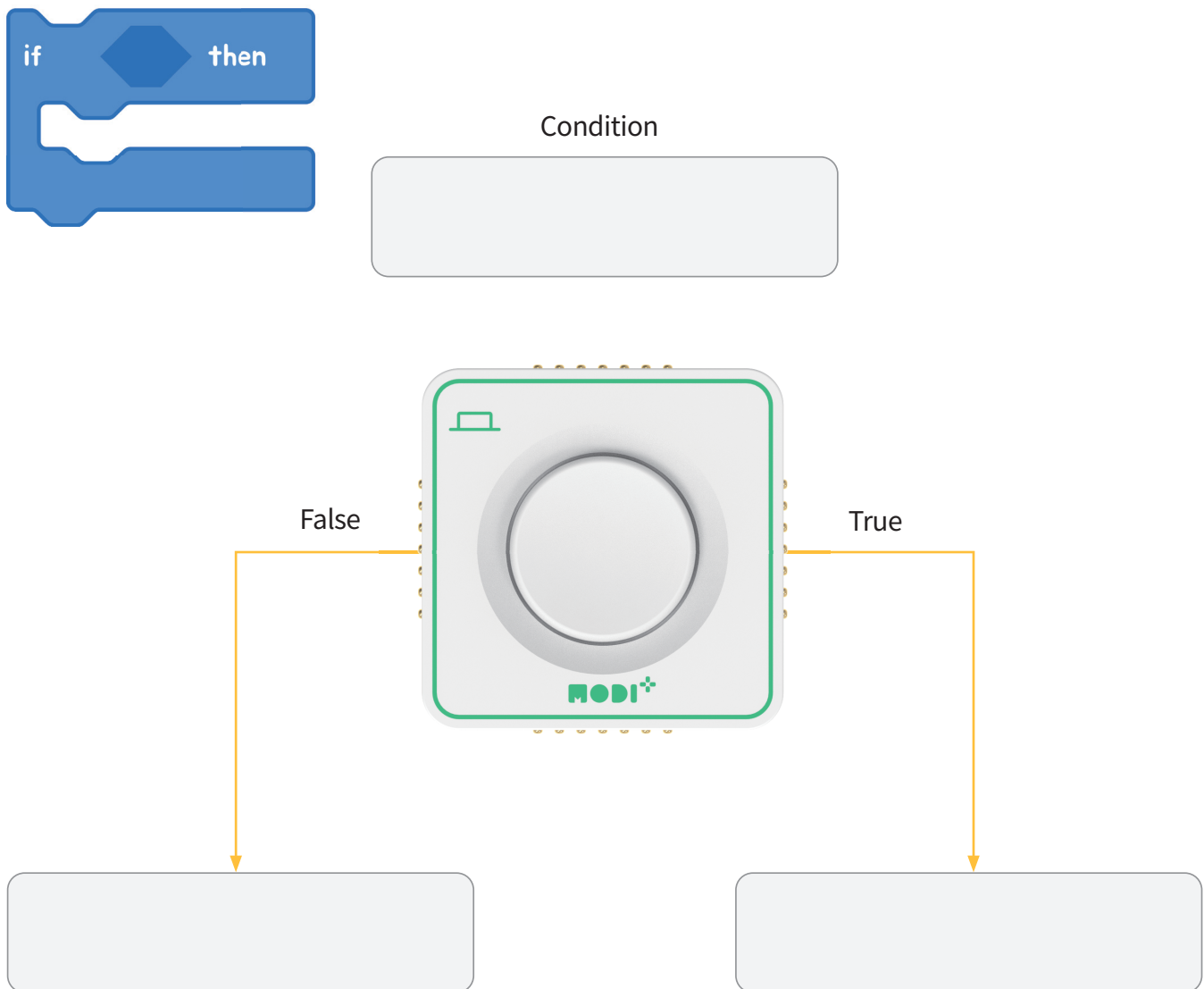


Conditional Statement2

Fill in the blanks with "condition," "false," and "true" as appropriate for the following statement:



Activity When the Button is clicked, the Speaker makes the alarm sound. If not, the Speaker is off.





IF ELSE STATEMENT



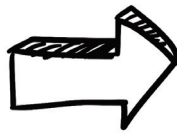
condition 1
If it's raining outside,



task 1
Take an umbrella.



condition 2
Otherwise,

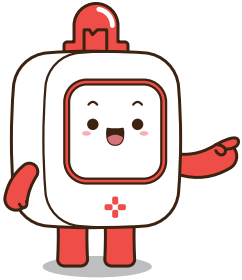


task 2
Leave it at home

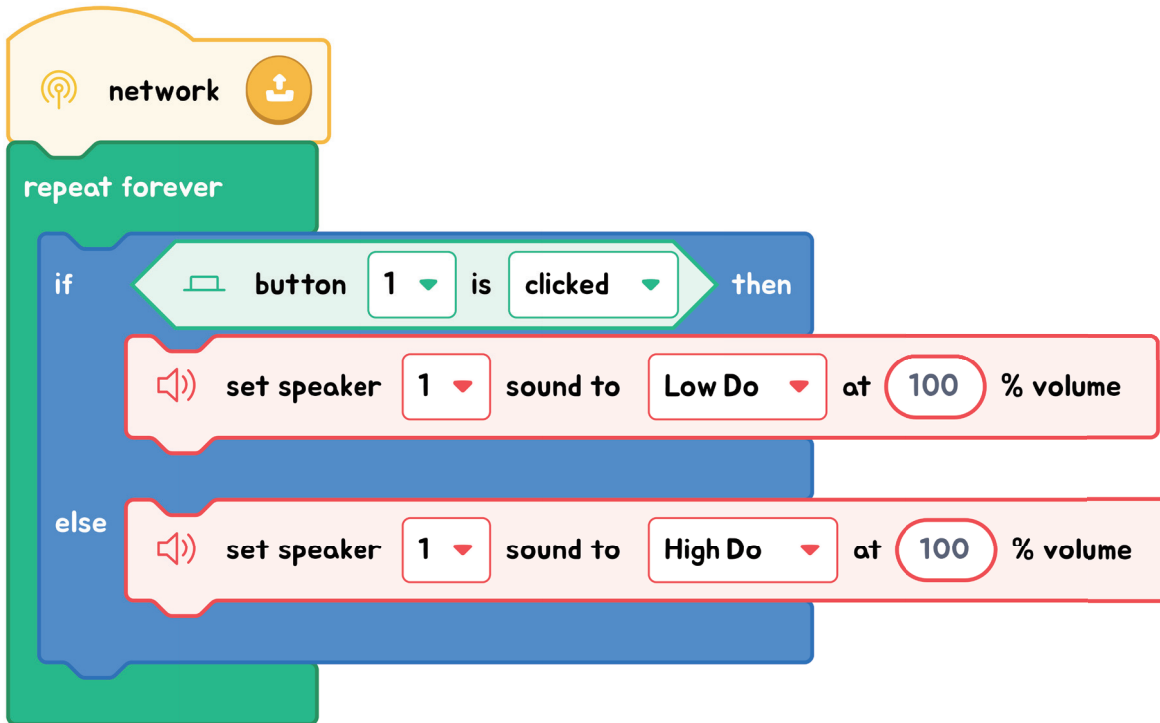
Imagine you have to decide whether to bring an umbrella outside. The rule could be: “If it's raining, take an umbrella. If it's not raining, you don't need to take an umbrella.”

The main point of the if-else statement is to execute the actions specified in the else part if the condition of the if statement is not met!





Here's a simple explanation and example:
Let's create the code like below and upload it!



Execution if False : While in most cases only the 'if' block is present, sometimes tasks can be specified to execute if the condition is false. Therefore, in this case, when the button is not clicked, the speaker makes the sound as High Do with 100% volume.

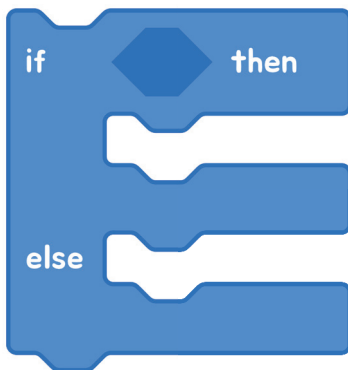


Conditional Statement 3

Fill in the blanks with "condition," "false," and "true" as appropriate for the following statement:



Activity When the Button is clicked, the Speaker makes the alarm sound and else the Speaker is off. If not, the Speaker is off.



Condition

False

True

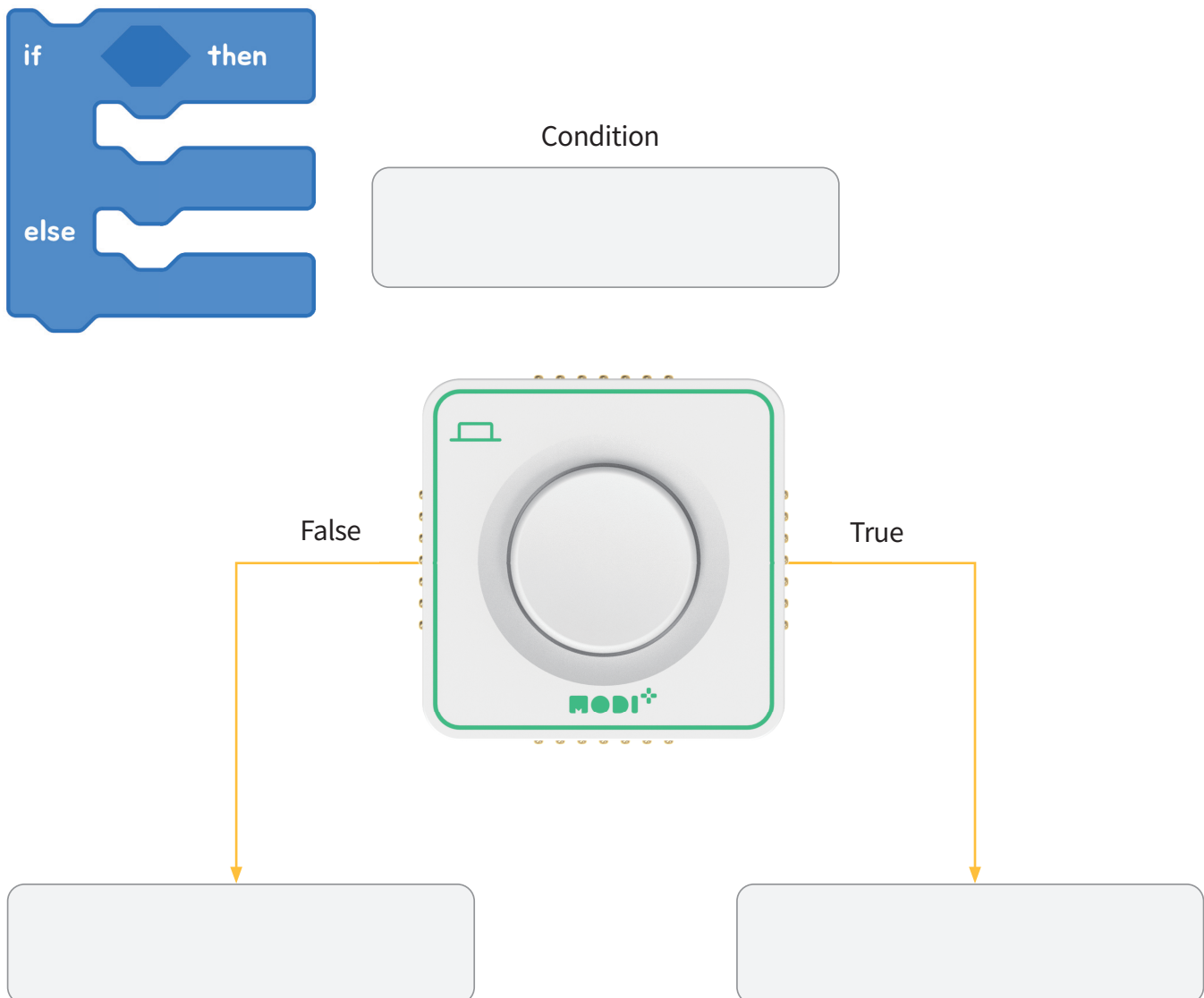


Conditional Statement 4

Fill in the blanks with "condition," "false," and "true" as appropriate for the following statement:



Activity If the Button is clicked, the LED will blink green 5 times. If not clicked, the LED will turn on with a red light.





● Multiple Choice Questions

1 What is programming?

- Ⓐ Telling stories to friends
- Ⓑ Giving instructions to a computer
- Ⓒ Drawing pictures
- Ⓓ Reading books

CORRECT ANSWER

Ⓑ

2 What happens when you use an "if" statement in programming?

- Ⓐ It repeats an action forever
- Ⓑ It checks a condition and makes a decision
- Ⓒ It stops the program
- Ⓓ It draws a picture

CORRECT ANSWER

Ⓑ

3 What does the "else" part of an "if else" statement do?

- Ⓐ It does nothing
- Ⓑ It checks another condition
- Ⓒ It tells the computer what to do if the "if" condition is false
- Ⓓ It starts a new program

CORRECT ANSWER

Ⓑ